

WMC_Ver03_Final_Model_Requirements

Author(s):Rakshith

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Requirement Set: WMC_Ver03_Final_Model_Requirements	2
1 System Initialization	3
2 Start Condition	3
3 Door Safety Before Start	3
4 Door Lock Activation	3
5 Wash-Cycle Water Fill	3
6 Detergent Fill Stage	4
7 Wash Stage	4
8 Wash-Cycle Drain	4
9 Rinse-Cycle Water Fill	5
10 Rinse Stage	5
11 Rinse-Cycle Drain	5
12 Spin Stage	5
13 Cycle Completion	6
14 Pause Command.....	6
15 Resume Command with History	6
16 Door Safety During Pause	6
17 Detergent Fill Timeout Fault.....	7
18 Generic Fault Detection	7
19 Fault Response.....	7
20 Fault Reset	7
21 Door Unlock at Safe States	8
22 Idle Output Reset.....	8
23 Actuator Mutual Exclusion.....	8
24 Stage Status Output	8
Artifact List.....	9

Requirement Set: WMC_Ver03_Final_Model_Requirements

Description

1 System Initialization

Requirement Type Functional

ID WMC-REQ-001

Description

On System start, the controller shall enter the Idle stage with all actuators OFF, cycle_complete set to 0, and fault_indicator set to 0.

Links

Artifact [WMC Ver 03 Final Model.slx](#)

 [Idle](#) (↔Implemented by)

2 Start Condition

Requirement Type Functional

ID WMC-REQ-002

Description

When start_button is pressed and door_closed is 1, the controller shall start the washing cycle.

Links

Artifact [WMC Ver 03 Final Model.slx](#)

 [\[start_button == 1 && door_closed == 1\]](#) (↔Implemented by)

3 Door Safety Before Start

Requirement Type Functional

ID WMC-REQ-003

Description

If start_button is pressed while door_closed is 0, the controller shall remain in Idle state and not initiate the cycle.

Links

Artifact [WMC Ver 03 Final Model.slx](#)

 [\[door_closed == 0\]](#) (↔Implemented by)

4 Door Lock Activation

Requirement Type Functional

ID WMC-REQ-004

Description

After a valid start command, the controller shall activate door_lock and shall wait for door_lock_delay duration before entering the Water_Fill_Wash_Cycle stage.

Links

Artifact [WMC Ver 03 Final Model.slx](#)

 [Door Lock Activation](#) (↔Implemented by)

 [\[after\(door_lock_delay, sec\) && door_closed == 1\]](#) (↔Implemented by)

5 Wash-Cycle Water Fill

Requirement Type Functional

ID WMC-REQ-005

Description

During Water_Fill_Wash_Cycle stage, the controller shall turn ON water_valve and shall remain in this stage until water_level_ok becomes 1. The water_level_ok is shared with the rinse-cycle water fill stage.

Links

Artifact [WMC Ver 03 Final Model.slx](#)

 [Water Fill Wash Cycle](#) (↔Implemented by)

 [\[water_level ok == 1\]](#) (↔Implemented by)

6 Detergent Fill Stage

Requirement Type Functional

ID WMC-REQ-006

Description

After wash-cycle water filling is completed, the controller shall enter Detergent_Fill stage and turn ON detergent_valve until detergent_level_ok becomes 1.

Links

Artifact [WMC Ver 03 Final Model.slx](#)

 [Detergent Fill](#) (↔Implemented by)

 [\[detergent_level ok == 1\]](#) (↔Implemented by)

7 Wash Stage

Requirement Type Functional

ID WMC-REQ-007

Description

After detergent filling is completed, the controller shall enter the Wash stage and turn on wash_motor for the configured wash_duration.

Links

Artifact [WMC Ver 03 Final Model.slx](#)

 [Wash](#) (↔Implemented by)

 [\[after\(wash_duration, sec\)\]](#) (↔Implemented by)

8 Wash-Cycle Drain

Requirement Type Functional

ID WMC-REQ-008

Description

After wash_duration is completed, the controller shall enter Drain_Wash_Cycle stage and turn ON drain_pump for the configured drain_duration. The drain_duration parameter is shared with the Drain_Rinse_Cycle stage.

Links

Artifact [WMC Ver 03 Final Model.slx](#)

 [Drain Wash Cycle](#) (↔Implemented by)

 [\[after\(drain_duration, sec\)\]](#) (↔Implemented by)

9 Rinse-Cycle Water Fill

Requirement Type Functional

ID WMC-REQ-009

Description

After wash-cycle draining is completed, the controller shall enter Water_Fill_Rinse_Cycle stage and shall turn ON water_valve until water_level_ok become 1. The water_level_ok signal is shared with the wash-cycle water fill stage.

Links

Artifact [WMC Ver 03 Final Model.slx](#)

 [Water Fill Rinse Cycle](#) (↔Implemented by)

 [\[water_level_ok == 1\]](#) (↔Implemented by)

10 Rinse Stage

Requirement Type Functional

ID WMC-REQ-010

Description

After rinse-cycle water filling is completed, the controller shall enter the Rinse stage and turn ON wash_motor for the configured rinse_duration.

Links

Artifact [WMC Ver 03 Final Model.slx](#)

 [Rinse](#) (↔Implemented by)

 [\[after\(rinse_duration, sec\)\]](#) (↔Implemented by)

11 Rinse-Cycle Drain

Requirement Type Functional

ID WMC-REQ-011

Description

After rinse_duration is completed, the controller shall enter Drain_Rinse_Cycle stage and turn ON drain_pump for the configured drain_duration. The drain_duration parameter is shared with the Drain_Wash_Cycle stage.

Links

Artifact [WMC Ver 03 Final Model.slx](#)

 [Drain Rinse Cycle](#) (↔Implemented by)

 [\[after\(drain_duration, sec\)\]](#) (↔Implemented by)

12 Spin Stage

Requirement Type Functional

ID WMC-REQ-012

Description

After rinse-cycle draining is completed, the controller shall enter Spin stage and turn ON the spin_motor for the configured spin_duration.

Links

Artifact [WMC Ver 03 Final Model.slx](#)

 [Spin](#) (↔Implemented by)

 [\[after\(spin_duration, sec\)\]](#) (↔Implemented by)

13 Cycle Completion

Requirement Type Functional

ID WMC-REQ-013

Description

After spin_duration is completed, the controller shall enter Cycle_Complete stage, turn OFF all actuators, shall unlock the door_lock, and shall set cycle_complete to 1.

Links

Artifact [WMC_Ver_03_Final_Model.slx](#)

 [Cycle_Complete](#) (↔Implemented by)

14 Pause Command

Requirement Type Functional

ID WMC-REQ-014

Description

If pause_button is pressed during any active cycle, the controller shall enter Paused stage and turn OFF active actuator outputs.

Links

Artifact [WMC_Ver_03_Final_Model.slx](#)

 [\[pause_button == 1\]](#) (↔Implemented by)

 [Paused](#) (↔Implemented by)

15 Resume Command with History

Requirement Type Functional

ID WMC-REQ-015

Description

If resume_button is pressed from the Paused stage and door_closed is 1, the controller shall resume from the previously interrupted cycle stage using Stateflow history junction behavior.

Links

Artifact [WMC_Ver_03_Final_Model.slx](#)

 [\[resume_button && door_closed == 1\]](#) (↔Implemented by)

 [Active Washing Cycle](#) (↔Implemented by)

16 Door Safety During Pause

Requirement Type Functional

ID WMC-REQ-016

Description

If door_closed is 0 during Paused stage, the controller shall not resume the cycle until door_closed becomes 1, even if resume_button is pressed.

Links

Artifact [WMC_Ver_03_Final_Model.slx](#)

 [\[resume button && door_closed == 1\]](#) (←Implemented by)

17 Detergent Fill Timeout Fault

Requirement Type Functional

ID WMC-REQ-017

Description

If detergent_level_ok does not become 1 within detergent_fill_timeout duration during Detergent_Fill stage, the controller shall enter Fault stage.

Links

Artifact [WMC Ver 03 Final Model.slx](#)

 [\[after\(detergent fill timeout, sec\) && detergent_level_ok == 0\]](#)

(←Implemented by)

18 Generic Fault Detection

Requirement Type Functional

ID WMC-REQ-018

Description

If fault_signal becomes 1 during any active cycle stage, the controller shall immediately transition to the Fault stage.

Links

Artifact [WMC Ver 03 Final Model.slx](#)

 [\[fault_signal == 1\]](#) (←Implemented by)

19 Fault Response

Requirement Type Functional

ID WMC-REQ-019

Description

In the Fault stage, the controller shall turn OFF water_valve, detergent_valve, drain_pump, wash_motor and spin_motor, and shall set fault_indicator to 1. door_lock shall remain ON in the Fault stage.

Links

Artifact [WMC Ver 03 Final Model.slx](#)

 [Fault](#) (←Implemented by)

20 Fault Reset

Requirement Type Functional

ID WMC-REQ-020

Description

If reset_button is pressed in the Fault stage and fault_signal is 0, the controller shall clear fault_indicator and with all outputs reset to Idle stage conditions.

Links

Artifact [WMC Ver 03 Final Model.slx](#)

 [\[reset_button == 1 & fault_signal == 0\]](#) (←Implemented by)

21 Door Unlock at Safe States

Requirement Type Functional

ID WMC-REQ-021

Description

The controller shall deactivate door_lock only in the following conditions: when entering Idle stage, when entering Cycle_Complete stage, or when valid fault reset occur (i.e. fault_signal is 0 AND reset_button is pressed in Fault stage).

Links

Artifact [WMC Ver 03 Final Model.slx](#)

 [Idle](#) (↔Implemented by)

 [Cycle Complete](#) (↔Implemented by)

 [\[reset_button == 1 & fault_signal == 0\]](#) (↔Implemented by)

22 Idle Output Reset

Requirement Type Functional

ID WMC-REQ-022

Description

In the Idle stage, all actuator outputs (water_valve, detergent_valve, drain_pump, wash_motor, spin_motor, door_lock) shall be OFF, cycle_complete shall be 0, and fault_indicator shall be 0.

Links

Artifact [WMC Ver 03 Final Model.slx](#)

 [Idle](#) (↔Implemented by)

23 Actuator Mutual Exclusion

Requirement Type Functional

ID WMC-REQ-023

Description

The controller shall ensure that wash_motor and spin_motor are never simultaneously ON at any point during operation.

Links

Artifact [WMC Ver 03 Final Model.slx](#)

 [Motor Mutual Exclusion Check](#) (↔Implemented by)

24 Stage Status Output

Requirement Type Functional

ID WMC-REQ-024

Description

The controller shall provide stage_status to indicate the currently active washing machine operating stage.

Links

Artifact [WMC Ver 03 Final Model.slx](#)

 [Stage Status Display](#) (↔Implemented by)

Appendix

Artifact List

Simulink models:

#	Name	Folder	Version
1	WMC_Ver_03_Final_Model.slx	D:\MBD-SILMIL_May_2026\MBD-Requirements_Based_Stateflow_Projects\Washing Machine Project\WMC_Ver03_Final_Model\model	2.42